1. Cycloidal Drive by MechFab3D

1.1 Printing instructions

It is recommended to print all parts in PETG or any other material that doesn't deform under load over time. For basic projects PLA is okay. Also print everything with 5 walls 25% infill and automatic tree support at 29°. Only the excentric shaft and the casing need support:

- Excentric shaft: print with the hole facing upwards away from the build plate. Print with automatic tree supports
- Cycloidal Disk: print with 4 walls
- Output Shaft 1: print with 3 or 4 walls to save a bit of filament since its not a structural part

You should tune your 3D Printer to be very precise if you haven't already. You can look up how to calibrate a 3D Printer with the X-Y hole and contour compensation settings.

Print every part one time except the cycloidal disks, which you need 2 times. Also, there are different sizes for fitting the bearings. You can choose a bigger or a smaller size if the bearings fit too loose or too tight.

Standard sizes are:

casing—9.9

cycloidal_disk—5.5_14.9 (1st number: outer holes, 2nd number: Inner Bearing Diameter)

excentric_shaft—9.9

lid—36.9

output_shaft_1—14.9

output_shaft_2—10_29.9 (1st Number: length, 2nd number: Outer Bearing Diameter)

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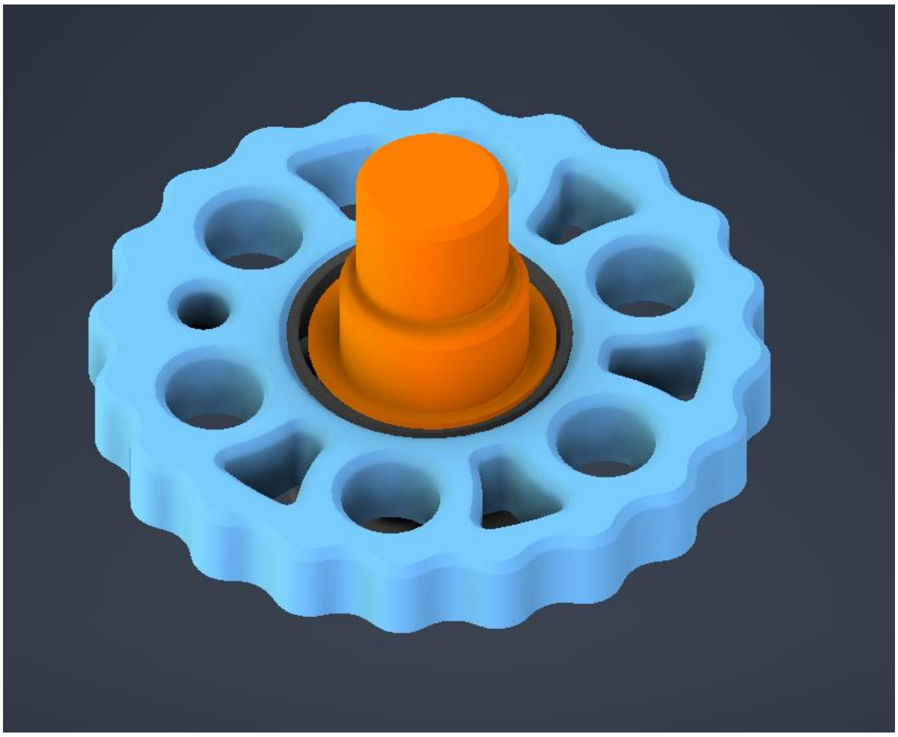
1.2 Assembly instructions

Info: There are multiple variations of every party with different diameters. Start printing with the standard version of every part. The first number of the cycloidal disk refers to the six outer holes' diameter. If the drive is hard to turn print the 5.6 version, if you have backlash problems print the 5.4 version. In case you have any questions with assembly or other issues with the drive, you can message me and we can work on a solution together.

1. Press the **10x15x4mm Bearings** into the middle of both **Cycloidal Disks**. You can flip the disk upside down and press down on it by using a flat surface like a table.

2. Press the **Cycloidal Disks** with the **Bearings** facing forwards onto the **Excentric Shaft.** The disks are flipped by 180° so that the sides, where the bearings were pressed in are on opposites sides and point to each other. Here is a single disk assembled. The other one needs to be flipped



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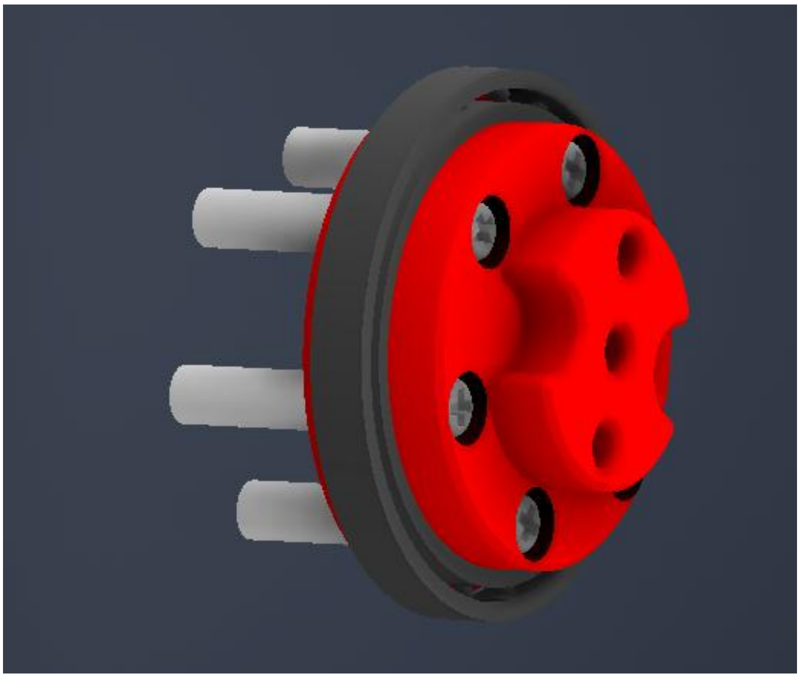
3. Press the **30x37x4mm Bearing** on **Output Shaft 2**
4. Put the six **M2.5x15mm Standoffs** in to **Output Shaft 2** on the opposite site of the bearing and screw them in with six **M2.5x8mm Screws**. You can hold the standoffs with a plier while screwing them in.
   Here is the assembled output shaft 2:



5. Take the disk and excentric shaft assembly and look for the **small holes** on both disk (there is a small hole in between the big holes). Use them to rotate both holes **on top of each other**. Here are the holes aligned marked in red (Note: a perfect alignment is not possible since they are not exactly on top of each other)

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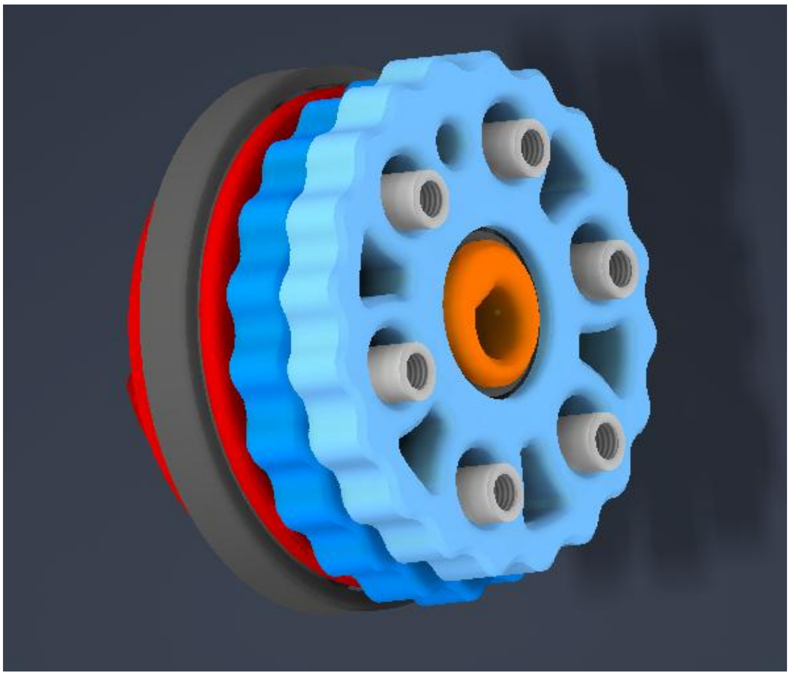
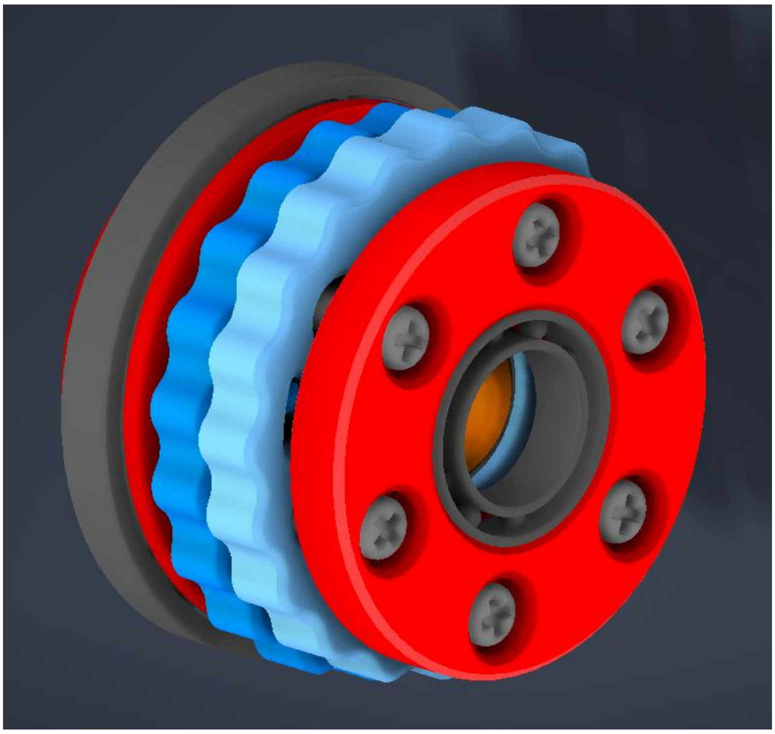
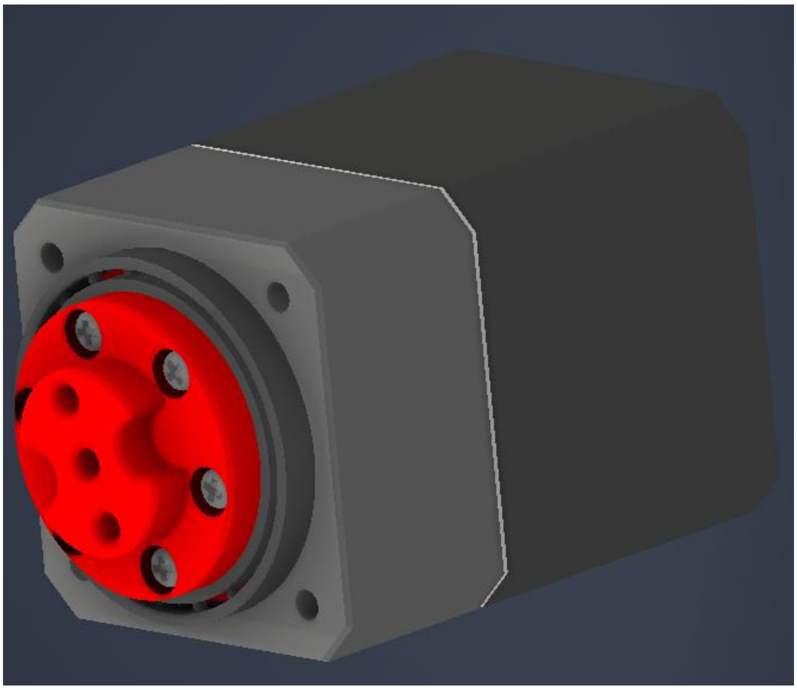
6. Take the **Output Shaft 2** with the standoffs and push them in to the aligned disks



7. Take the **Output Shaft 1** and put it on the assembly like a lid and screw it in with six **M2.5x8mm screw**
8. Press another **10x15x4mm Bearing** into the middle hole of **Output Shaft 1**. The bearing must be facing towards the outside, otherwise **Output Shaft 1** needs to be flipped

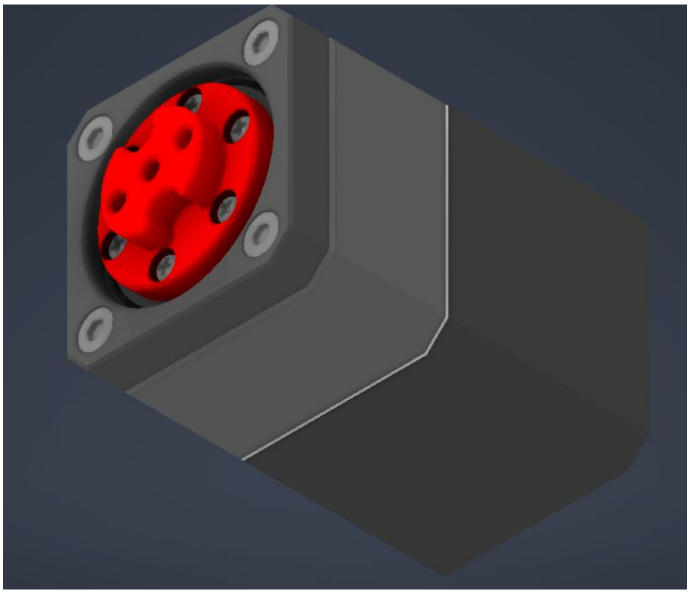


9. Now the whole inner assembly can be put into the case with the small bearing first. You must rotate the assembly a bit, and it should slide in. If the fit is too tight or too lose you might need to use another size of the cycloidal disks. (Use the test drive shaft to spin and test the drive). Now you can put everything on to the NEMA 17



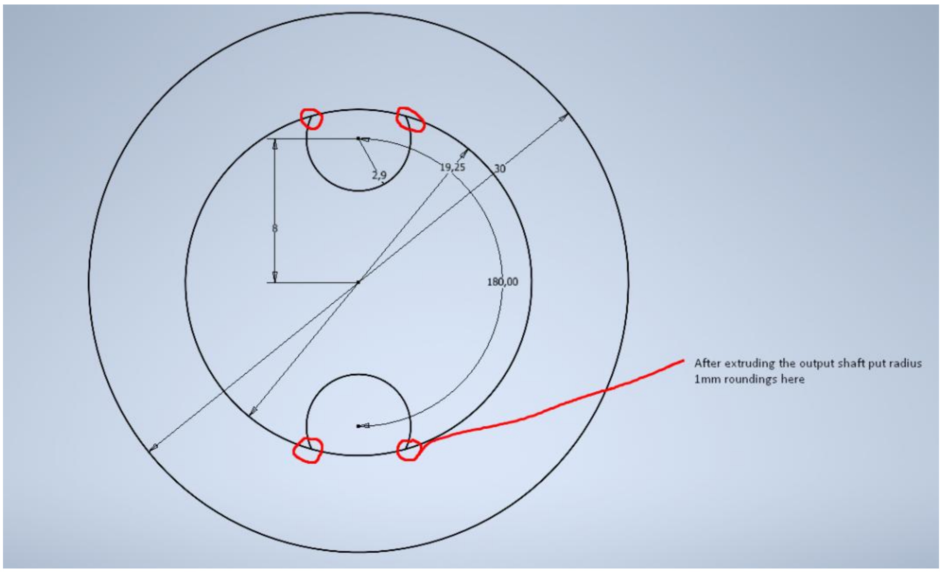
10. Take the **Lid** and press it on the **30x37x4mm Bearing** and screw in the four **M3x30mm Screws**

11. Put two **M3 Heat Inserts** into outer holes of **Output Shaft 2**



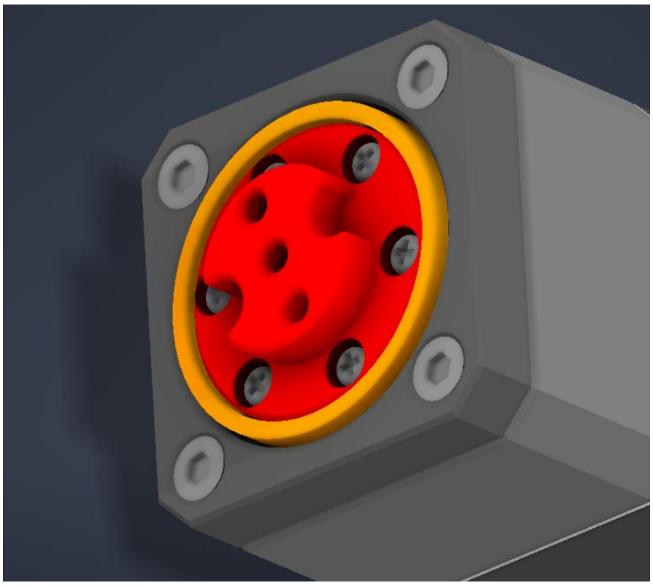
1.3 Inclusion in your project

To use the drive, you can use my design of the output shaft. The sketch shows the geometry your shaft should have to slide smoothly on the output shaft of the cycloidal drive.

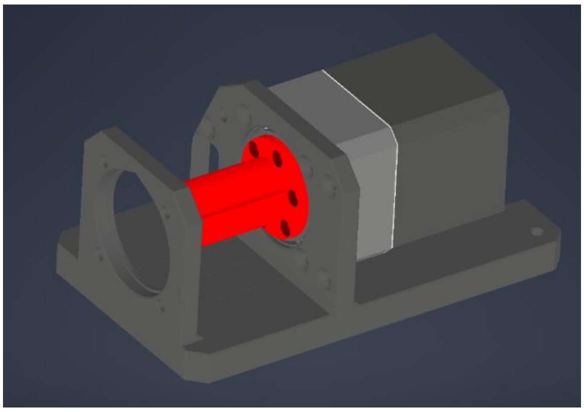


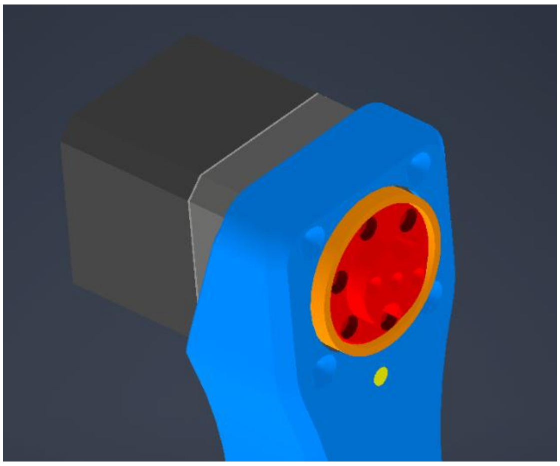
I have made 3 different versions of the output shaft with profiles. You can choose between these or message me and I can model you a custom output shaft.

There is a distance ring included which you should put onto the output shaft 2. It reduces the vertical movement when a shaft is connected to the drive (here in orange)



For my projects i use the lid and model the mounting mechanism i need around it. In the end I put the assembly into the lid and screw it in like so:





I include the step file for the lid so you can remodel it for your project.

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1.4 Parts needed

All the parts you need

Note: The links are affiliate links so I get a small promotion if you buy with the link. You don't have to pay more for the product, but you would support me and my projects. Thank you very much

Part	Type	Amount	Link
Bearing	10x15x4	3	https://s.click.aliexpress.com/e/_c2uAe96f
Bearing	30x37x4	1	https://s.click.aliexpress.com/e/_c2uAe96f
Standoff	M2.5x15	6	https://s.click.aliexpress.com/e/_c4mglond
Philipps screw	M2.5x8	12	https://s.click.aliexpress.com/e/_c4DnXWhh
Countersunk screw	M3x30	4	https://s.click.aliexpress.com/e/_c30eY4qx
Heat Inserts	M3x4	2	https://s.click.aliexpress.com/e/_c35Fsfd9

Here are some additional parts i use to make the code work:

Part	Type	Amount	Link
Stepper motor	NEMA 17	1	https://s.click.aliexpress.com/e/_c40i0GEj You can choose different sizes depending on your needs
Microcontroller	ESP32	1	https://s.click.aliexpress.com/e/_c3PJD7vl Get the CP2102 version since its a better chip than the CH340
Motor driver	TMC2209	1	https://s.click.aliexpress.com/e/_c3yualpZ
Motor driver	MKS Servo42c	1	https://s.click.aliexpress.com/e/_c3RLFAST This driver sits directly on the motor and has a closed loop system. Its probably overkill for most people but its also very good
Cables			https://s.click.aliexpress.com/e/_c4FbWWMf

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1.5 Disclaimer

This model and guide are provided as-is without any warranty.

Use at your own risk. MechFab3D assumes no responsibility for damages or injuries resulting from the use of these files.

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